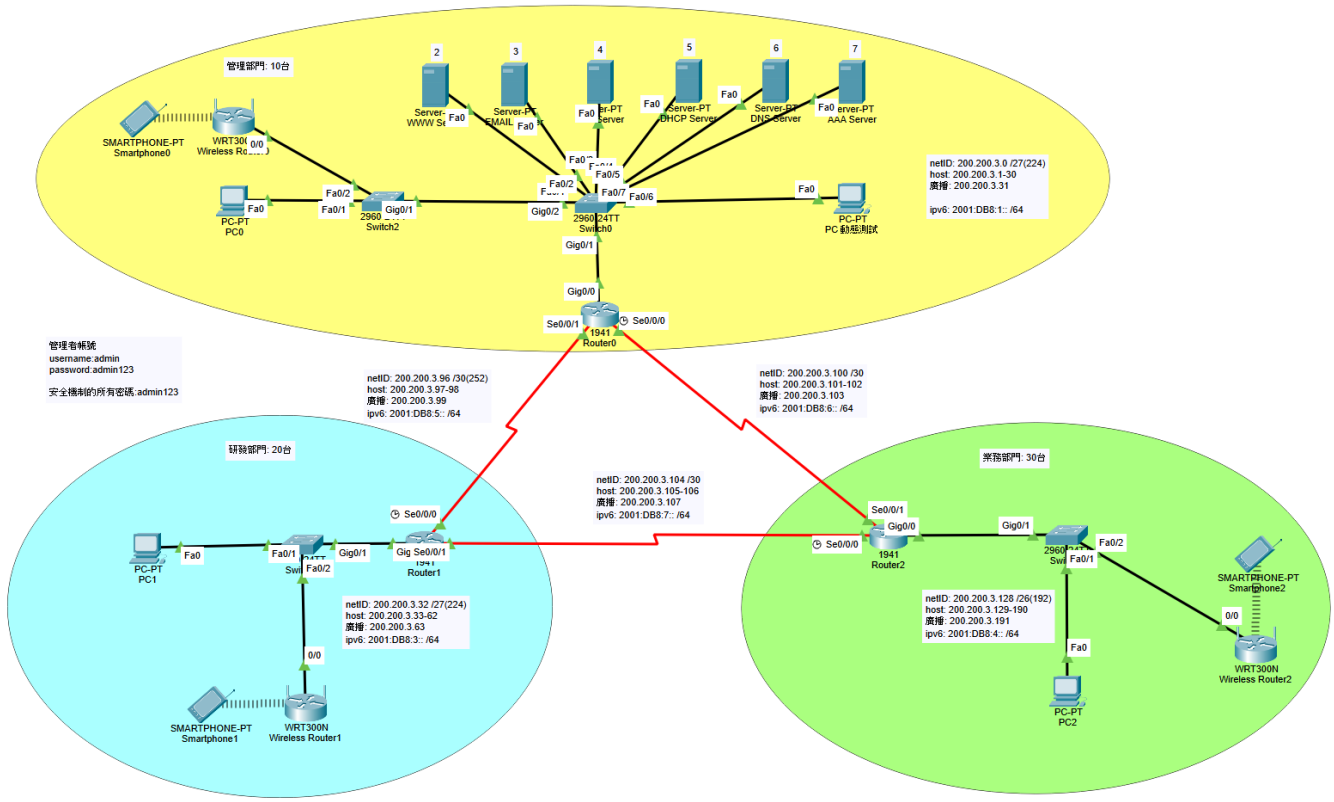
1. 該公司共有 3 個部門，管理部門 10 人、研發部門 20 人與業務部門 30 人
2. IP 位址：200.200.XX.0 /24 (VLSM)



3. Dual Stack (整合 IPv4 與 IPv6 網路)

PC1 ping PC0 (IPv4)

```
C:\>ping 200.200.3.12

Pinging 200.200.3.12 with 32 bytes of data:

Reply from 200.200.3.12: bytes=32 time=18ms TTL=126
Reply from 200.200.3.12: bytes=32 time=15ms TTL=126
Reply from 200.200.3.12: bytes=32 time=1ms TTL=126
Reply from 200.200.3.12: bytes=32 time=1ms TTL=126

Ping statistics for 200.200.3.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 18ms, Average = 8ms
```

PC1 ping PC0 (IPv6)

```
C:\>ping 2001:DB8:1::9

Pinging 2001:DB8:1::9 with 32 bytes of data:

Reply from 2001:DB8:1::9: bytes=32 time=20ms TTL=126
Reply from 2001:DB8:1::9: bytes=32 time=25ms TTL=126
Reply from 2001:DB8:1::9: bytes=32 time=15ms TTL=126
Reply from 2001:DB8:1::9: bytes=32 time=2ms TTL=126

Ping statistics for 2001:DB8:1::9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 25ms, Average = 15ms
```

PC1 ping PC2 (IPv4)

```
C:\>ping 200.200.3.130

Pinging 200.200.3.130 with 32 bytes of data:

Reply from 200.200.3.130: bytes=32 time=33ms TTL=126
Reply from 200.200.3.130: bytes=32 time=23ms TTL=126
Reply from 200.200.3.130: bytes=32 time=24ms TTL=126
Reply from 200.200.3.130: bytes=32 time=1ms TTL=126

Ping statistics for 200.200.3.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 33ms, Average = 20ms
```

PC1 ping PC2 (IPv6)

```
C:\>ping 2001:DB8:4::2

Pinging 2001:DB8:4::2 with 32 bytes of data:

Reply from 2001:DB8:4::2: bytes=32 time=12ms TTL=126
Reply from 2001:DB8:4::2: bytes=32 time=2ms TTL=126
Reply from 2001:DB8:4::2: bytes=32 time=2ms TTL=126
Reply from 2001:DB8:4::2: bytes=32 time=2ms TTL=126

Ping statistics for 2001:DB8:4::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 4ms
```

PC1 tracert PC0 & PC2

```
C:\>tracert 200.200.3.130

Tracing route to 200.200.3.130 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      200.200.3.33
  2    17 ms     0 ms      0 ms      200.200.3.106
  3    0 ms      0 ms      1 ms      200.200.3.130

Trace complete.

C:\>tracert 200.200.3.12

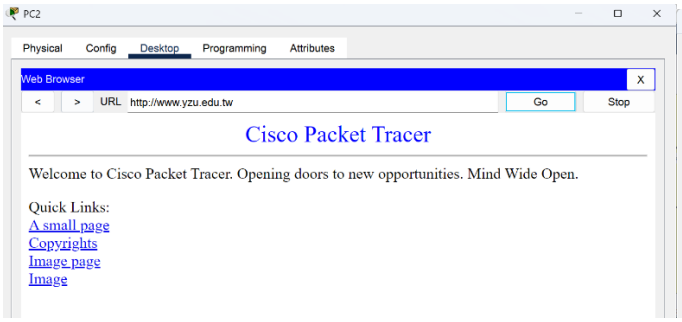
Tracing route to 200.200.3.12 over a maximum of 30 hops:

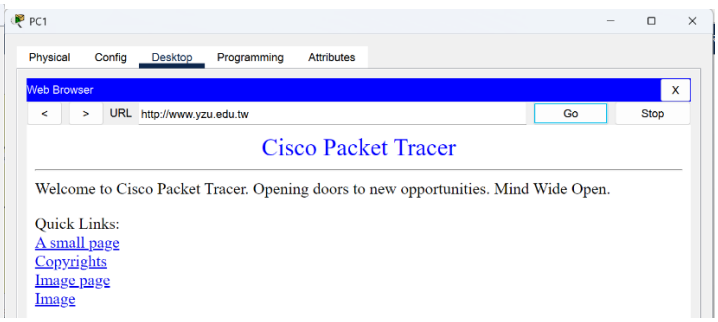
  1    0 ms      0 ms      0 ms      200.200.3.33
  2    0 ms      2 ms      5 ms      200.200.3.98
  3    1 ms      0 ms      1 ms      200.200.3.12

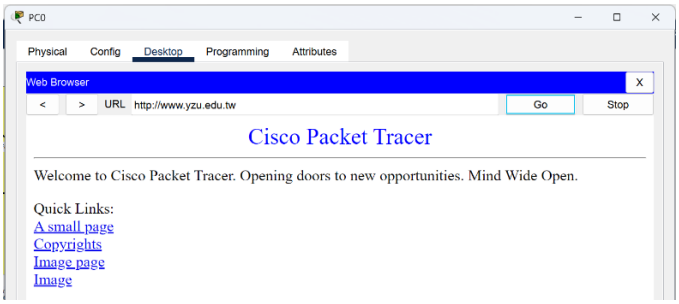
Trace complete.
```

4. 該公司內部提供 Web、Email、FTP、DHCP、DNS 與 Wi-Fi 服務(含 AAA Server)

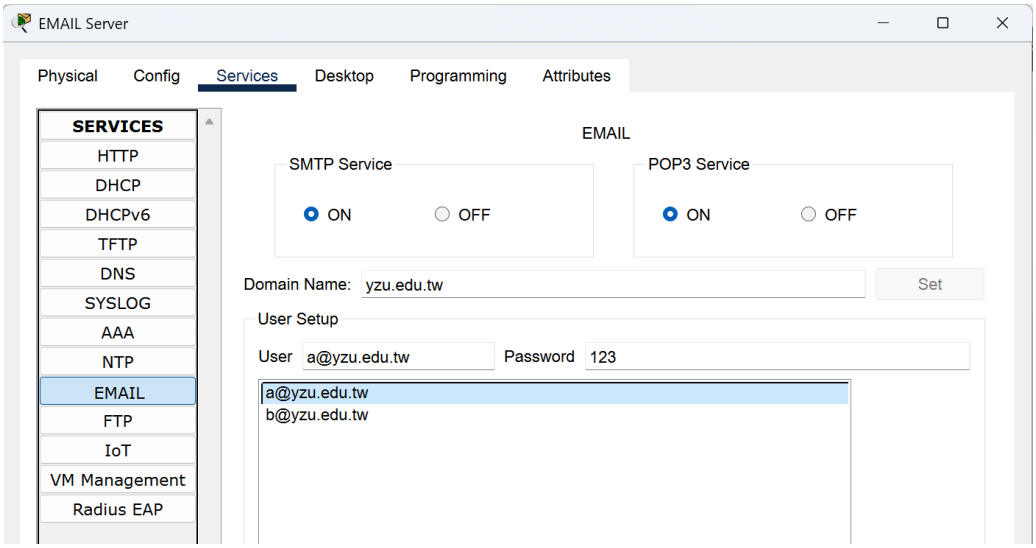
Web:

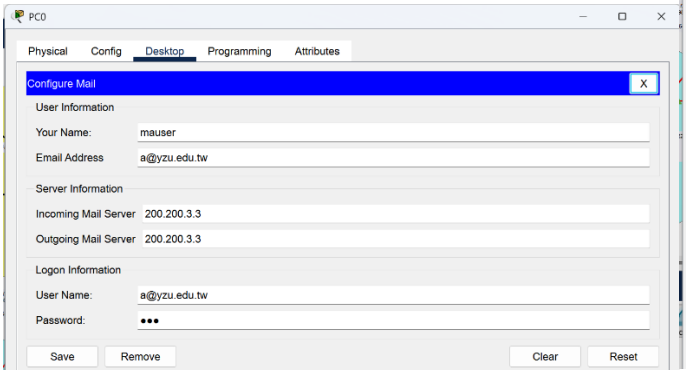


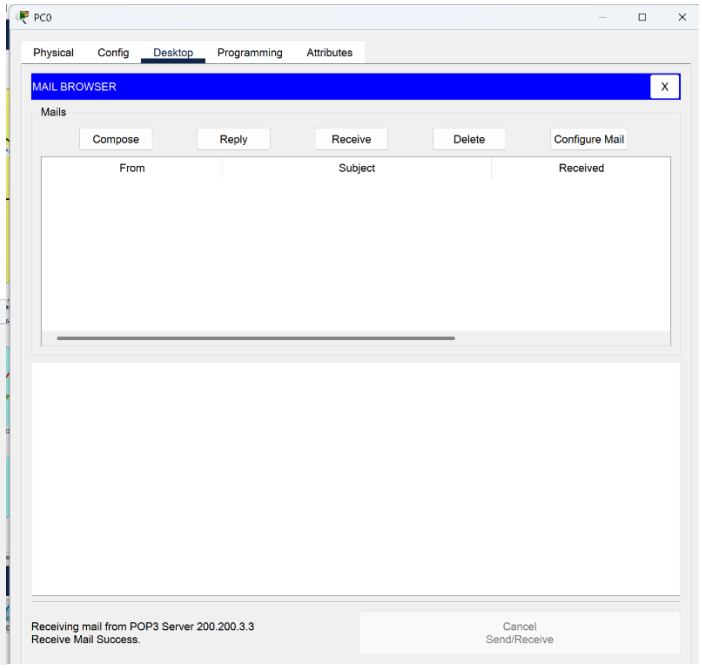



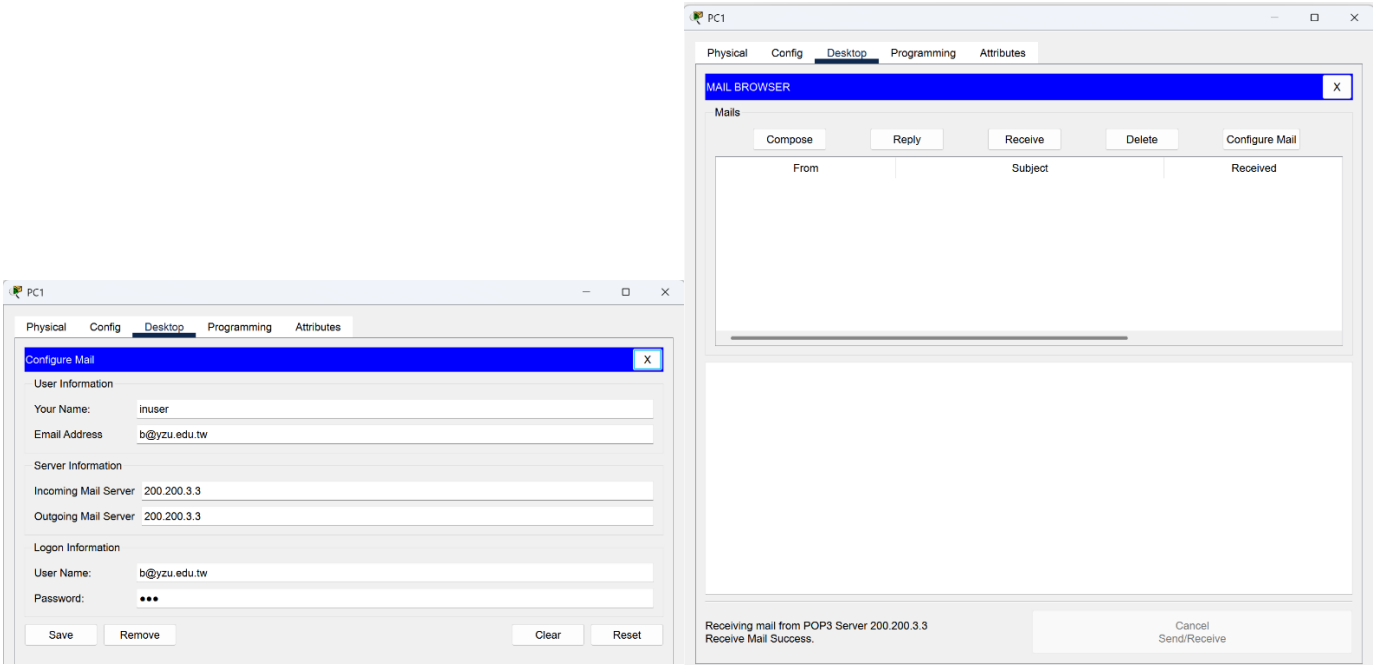




Email:

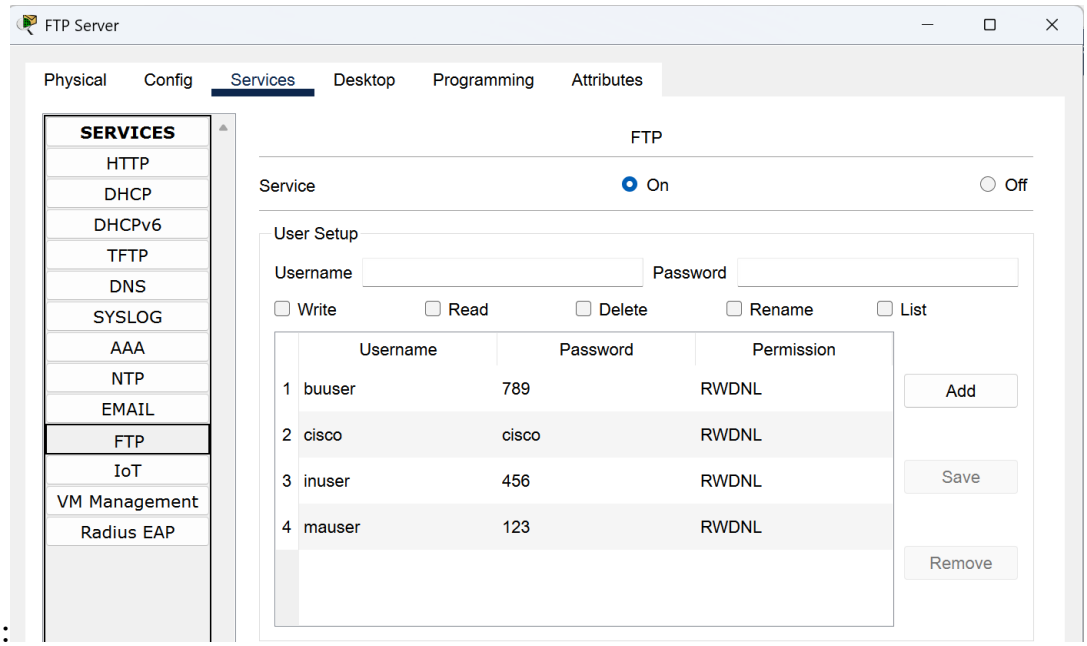





FTP:




```
C:\>ftp 200.200.3.4
Trying to connect...200.200.3.4
Connected to 200.200.3.4
220- Welcome to PT Ftp server
Username:mauser
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

```
C:\>ftp 200.200.3.4
Trying to connect...200.200.3.4
Connected to 200.200.3.4
220- Welcome to PT Ftp server
Username:inuser
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ftp 200.200.3.4
Trying to connect...200.200.3.4
Connected to 200.200.3.4
220- Welcome to PT Ftp server
Username:buuser
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

DHCP:

DHCP Server

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP

DHCP

| | | | | |
|---|---|---|---|---|
| Interface | FastEthernet0 | Service | On | Off |
| Pool Name | serverPool | | | |
| Default Gateway | 200.200.3.1 | | | |
| DNS Server | 200.200.3.6 | | | |
| Start IP Address : | 200 | 200 | 3 | 10 |
| Subnet Mask: | 255 | 255 | 255 | 224 |
| Maximum Number of Users : | 11 | | | |
| TFTP Server: | 0.0.0.0 | | | |
| WLC Address: | 0.0.0.0 | | | |

Add | Save | Remove

| Pool Name | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|---|---|---|---|---|---|---|---|
| Bu-Pool | 200.200.3.129 | 200.200.3.6 | 200.200.3.130 | 255.255.255... | 30 | 0.0.0.0 | 0.0.0.0 |
| serverPool | 200.200.3.1 | 200.200.3.6 | 200.200.3.10 | 255.255.255... | 11 | 0.0.0.0 | 0.0.0.0 |
| In-Pool | 200.200.3.33 | 200.200.3.6 | 200.200.3.34 | 255.255.255... | 20 | 0.0.0.0 | 0.0.0.0 |

IP Configuration

DHCP | Static

| | |
|---|---|
| IPv4 Address | 200.200.3.12 |
| Subnet Mask | 255.255.255.224 |
| Default Gateway | 200.200.3.1 |
| DNS Server | 200.200.3.6 |

IP Configuration

DHCP | Static

| | |
|---|---|
| IPv4 Address | 200.200.3.130 |
| Subnet Mask | 255.255.255.192 |
| Default Gateway | 200.200.3.129 |
| DNS Server | 200.200.3.6 |

PC1

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

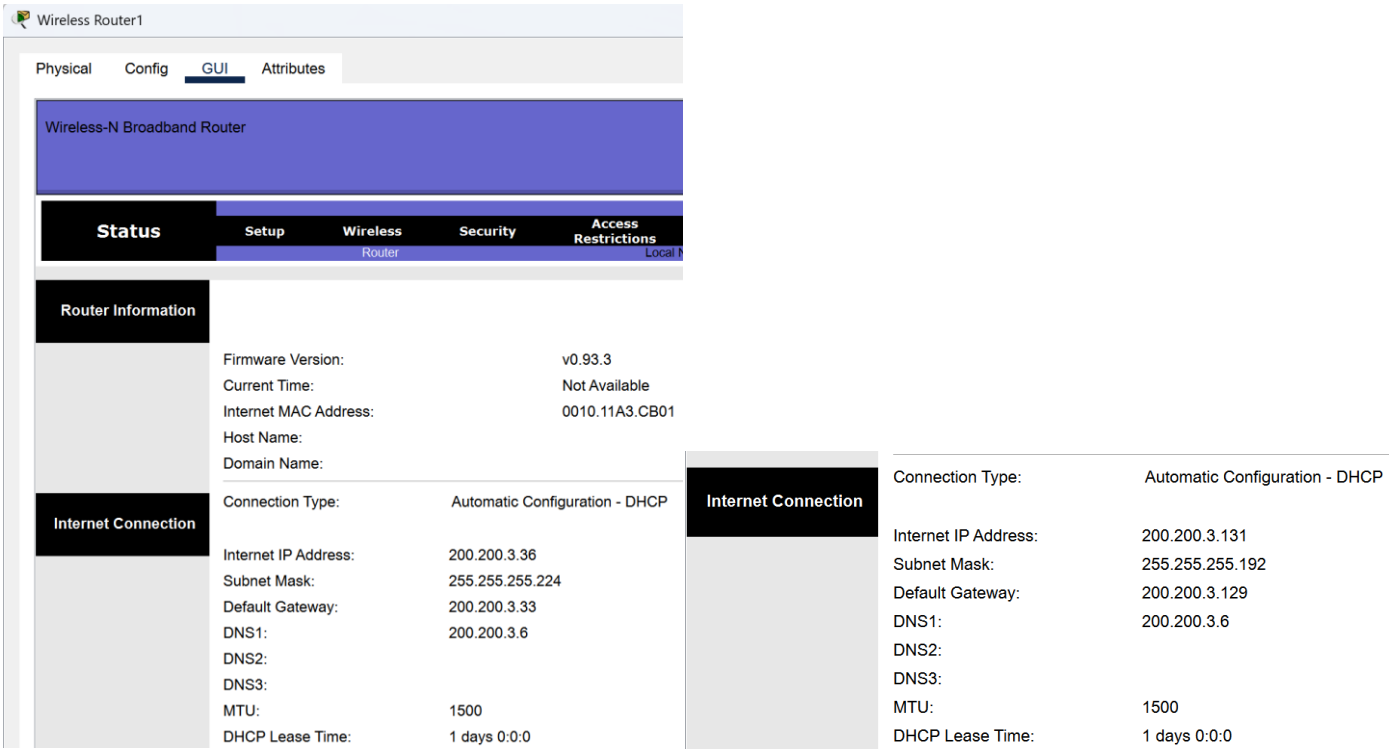
DHCP | Static

| | |
|---|---|
| IPv4 Address | 200.200.3.35 |
| Subnet Mask | 255.255.255.224 |
| Default Gateway | 200.200.3.33 |
| DNS Server | 200.200.3.6 |

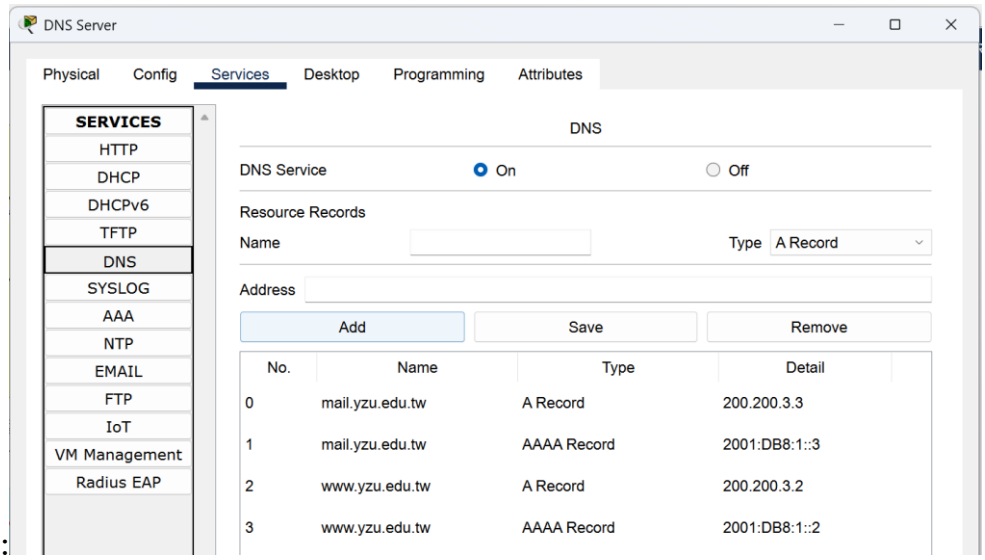
Internet Connection

| | |
|---|---|
| Connection Type: | Automatic Configuration - DHCP |
| Internet IP Address: | 200.200.3.14 |
| Subnet Mask: | 255.255.255.224 |
| Default Gateway: | 200.200.3.1 |
| DNS1: | 200.200.3.6 |
| DNS2: | |
| DNS3: | |
| MTU: | 1500 |
| DHCP Lease Time: | 1 days 0:0:0 |

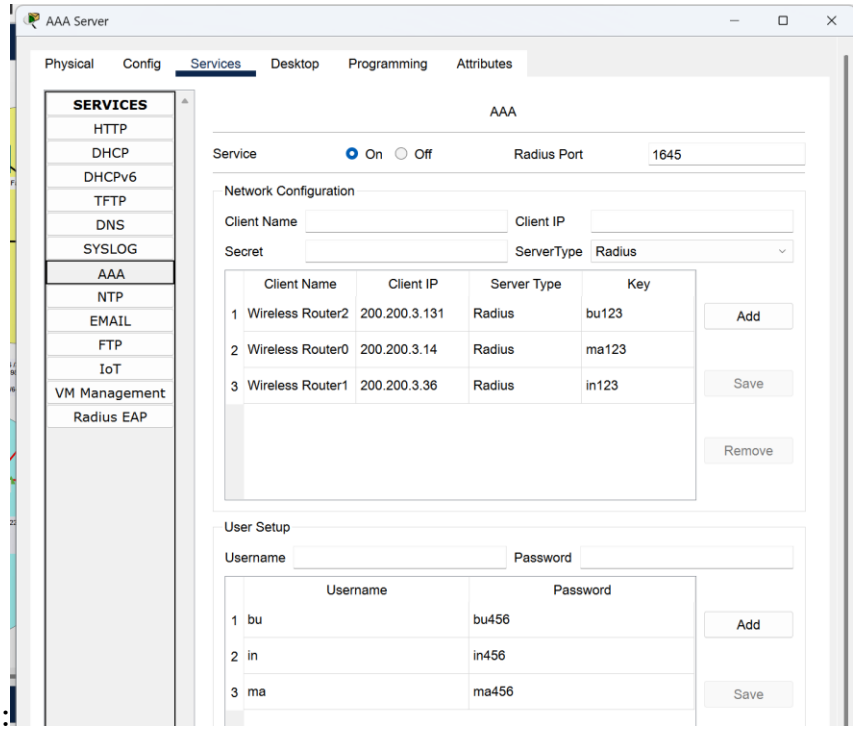
IP Address Release | IP Address Renew

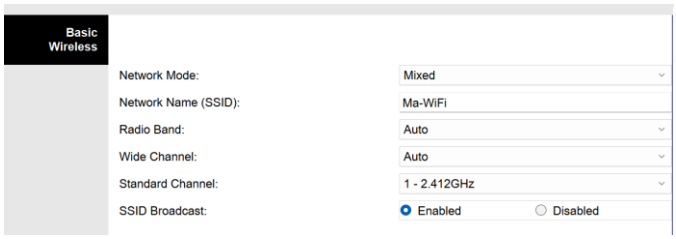

Wireless Router1
Physical
Config
GUI
Attributes
Wireless-N Broadband Router
Status
Setup
Wireless
Security
Access Restrictions
Router
Router Information
Firmware Version: v0.93.3
Current Time: Not Available
Internet MAC Address: 0010.11A3.CB01
Host Name:
Domain Name:
Internet Connection
Connection Type: Automatic Configuration - DHCP
Internet IP Address: 200.200.3.36
Subnet Mask: 255.255.255.224
Default Gateway: 200.200.3.33
DNS1: 200.200.3.6
DNS2:
DNS3:
MTU: 1500
DHCP Lease Time: 1 days 0:0:0
Internet Connection
Connection Type: Automatic Configuration - DHCP
Internet IP Address: 200.200.3.131
Subnet Mask: 255.255.255.192
Default Gateway: 200.200.3.129
DNS1: 200.200.3.6
DNS2:
DNS3:
MTU: 1500
DHCP Lease Time: 1 days 0:0:0

DNS:

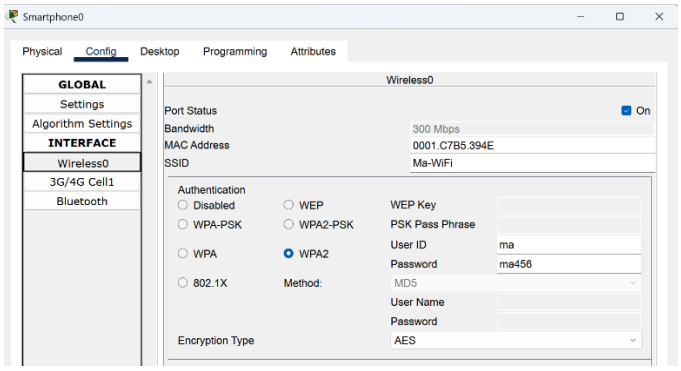

DNS Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0 mail.yzu.edu.tw A Record 200.200.3.3
1 mail.yzu.edu.tw AAAA Record 2001:DB8:1::3
2 www.yzu.edu.tw A Record 200.200.3.2
3 www.yzu.edu.tw AAAA Record 2001:DB8:1::2

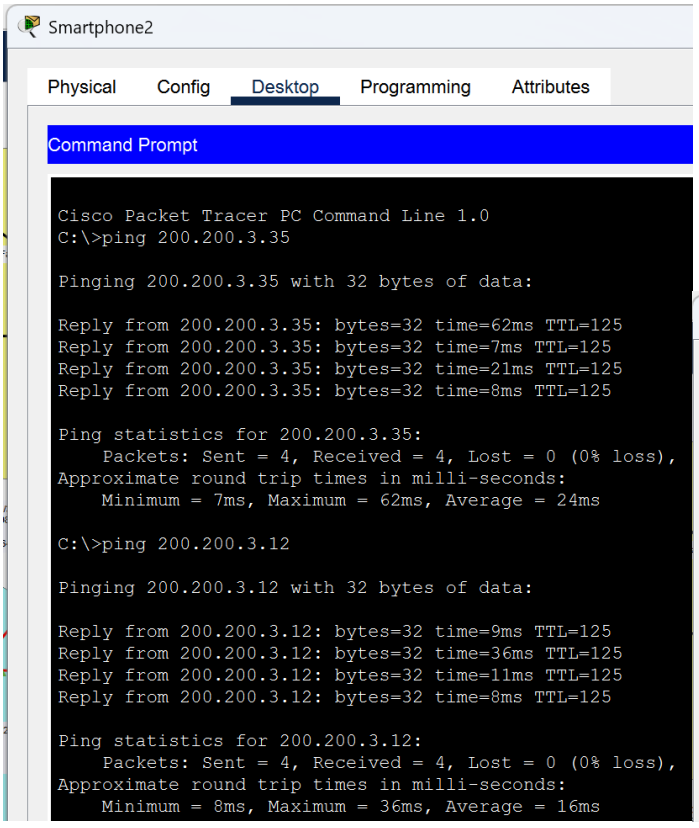
Wi-Fi(AAA):

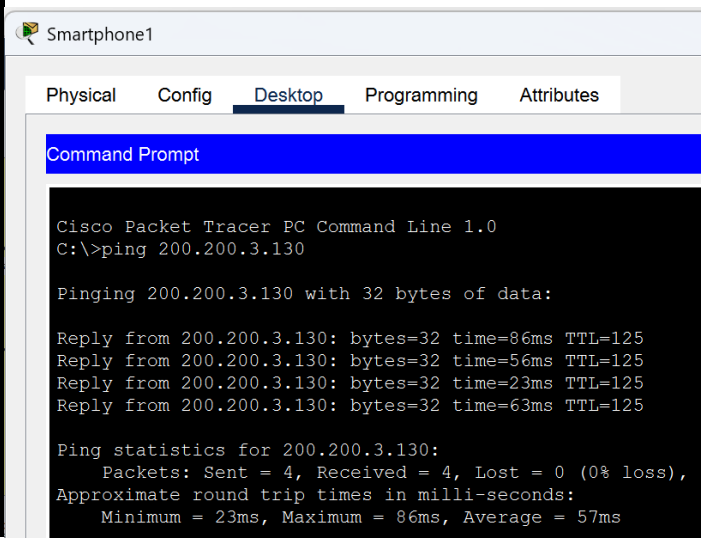

AAA Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
AAA
Service
On
Off
Radius Port
1645
Network Configuration
Client Name
Client IP
Secret
ServerType
Radius
Client Name
Client IP
Server Type
Key
1 Wireless Router2 200.200.3.131 Radius bu123
2 Wireless Router0 200.200.3.14 Radius ma123
3 Wireless Router1 200.200.3.36 Radius in123
Add
Save
Remove
User Setup
Username
Password
Username
Password
1 bu bu456
2 in in456
3 ma ma456
Add
Save


Basic Wireless
Network Mode: Mixed
Network Name (SSID): Ma-WiFi
Radio Band: Auto
Wide Channel: Auto
Standard Channel: 1 - 2.412GHz
SSID Broadcast: Enabled Disabled


Wireless Security
Security Mode: WPA2 Enterprise
Encryption: AES
RADIUS Server: 200 . 200 . 3 . 7
RADIUS Port: 1645
Shared Secret: ma123
Key Renewal: 3600 seconds


Smartphone0
Physical Config Desktop Programming Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
3G/4G Cell1
Bluetooth
Wireless0
Port Status On
Bandwidth 300 Mbps
MAC Address 0001.C7B5.394E
SSID Ma-WiFi
Authentication
Disabled WEP
WPA-PSK WPA2-PSK
WPA WPA2
802.1X
WEP Key
PSK Pass Phrase
User ID ma
Password ma456
Method: MD5
User Name
Password
Encryption Type AES


Smartphone2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.200.3.35
Pinging 200.200.3.35 with 32 bytes of data:
Reply from 200.200.3.35: bytes=32 time=62ms TTL=125
Reply from 200.200.3.35: bytes=32 time=7ms TTL=125
Reply from 200.200.3.35: bytes=32 time=21ms TTL=125
Reply from 200.200.3.35: bytes=32 time=8ms TTL=125
Ping statistics for 200.200.3.35:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 7ms, Maximum = 62ms, Average = 24ms
C:\>ping 200.200.3.12
Pinging 200.200.3.12 with 32 bytes of data:
Reply from 200.200.3.12: bytes=32 time=9ms TTL=125
Reply from 200.200.3.12: bytes=32 time=36ms TTL=125
Reply from 200.200.3.12: bytes=32 time=11ms TTL=125
Reply from 200.200.3.12: bytes=32 time=8ms TTL=125
Ping statistics for 200.200.3.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 36ms, Average = 16ms


Smartphone1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.200.3.130
Pinging 200.200.3.130 with 32 bytes of data:
Reply from 200.200.3.130: bytes=32 time=86ms TTL=125
Reply from 200.200.3.130: bytes=32 time=56ms TTL=125
Reply from 200.200.3.130: bytes=32 time=23ms TTL=125
Reply from 200.200.3.130: bytes=32 time=63ms TTL=125
Ping statistics for 200.200.3.130:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 23ms, Maximum = 86ms, Average = 57ms

5. 請設定啟動所有網路設備(Router, Switch, and Wireless Router)之安全機制

```
User Access Verification

Password:

R0>en
Password:
R0#show running-config
Building configuration...

Current configuration : 1550 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
security passwords min-length 8
!
hostname R0
!
login block-for 60 attempts 3 within 120
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
username admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
!
license udi pid CISCO1941/K9 sn FTX15248N80-
line con 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 exec-timeout 5 30
 password 7 082048430017544541
 login local
 transport input ssh
```